

TRUBCHEVSK, Russian Federation

WMO: 269970

Lat: 52.583N

Lon: 33.767E

Elev: 178

StdP: 99.20

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.1	-25.0	0.4	-22.3	-21.7	0.5	-18.9	10.1	-1.2	8.4	-3.3	1.3	30	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.1	20.4	28.3	19.6	26.6	18.9	21.7	27.3	20.8	26.3	19.9	25.2	2.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.9	24.9	18.9	14.0	23.6	18.0	13.2	22.4	64.3	27.3	60.9	26.3	57.7	25.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.3	7.0	5.7	DB	-25.4	32.5	3.2	2.0	-27.8	34.0	-29.7	35.1	-31.5	36.2	-33.8	37.7	
			WB	-25.6	23.5	3.2	1.4	-27.9	24.5	-29.8	25.3	-31.6	26.1	-33.9	27.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-5.8	-5.2	-0.1	8.1	14.2	17.5	19.6	18.2	12.7	6.6	0.9	-3.8	(6)
(7)		DBStd	10.25	6.03	6.38	4.66	4.39	4.53	3.57	3.08	3.48	3.96	4.24	4.99	5.96	(7)
(8)		HDD10.0	2175	490	426	313	87	14	1	0	0	18	123	274	429	(8)
(9)		HDD18.3	4300	748	659	571	306	142	57	22	46	172	365	523	688	(9)
(10)		CDD10.0	1068	0	0	1	31	144	227	297	253	99	16	1	0	(10)
(11)		CDD18.3	152	0	0	0	0	13	33	61	41	4	0	0	0	(11)
(12)		CDH23.3	1412	0	0	0	4	143	282	509	422	51	1	0	0	(12)
(13)		CDH26.7	372	0	0	0	1	25	56	148	138	5	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.6	2.6	2.7	2.5	2.3	2.2	1.9	1.9	2.0	2.3	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	614	41	36	34	39	61	67	81	56	58	54	47	40	(15)
(16)		PrecMax	793	80	68	80	97	105	173	172	133	125	126	91	84	(16)
(17)		PrecMin	422	10	16	6	3	18	10	10	1	6	6	1	13	(17)
(18)		PrecStd	85	15	14	16	21	20	31	41	34	28	31	22	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.7	14.7	23.4	29.2	30.2	32.8	34.0	27.2	20.9	13.4	8.3	(19)
(20)		MCWB	4.4	4.0	8.6	12.9	19.1	20.1	21.1	20.0	17.3	14.5	11.6	7.3	(20)	
(21)		2%	DB	3.2	3.7	11.1	20.6	26.9	28.3	30.2	30.1	24.9	17.3	10.3	6.1	(21)
(22)		MCWB	2.4	2.5	6.3	12.2	17.9	19.6	20.7	20.2	17.0	13.4	9.0	5.3	(22)	
(23)		5%	DB	2.1	2.6	8.3	18.2	24.8	26.5	28.1	27.6	21.9	14.7	8.5	3.8	(23)
(24)		MCWB	1.4	1.7	4.6	11.3	17.0	18.8	20.3	19.5	15.6	11.9	7.5	3.2	(24)	
(25)		10%	DB	1.0	1.8	6.0	15.8	22.5	24.7	26.2	25.3	19.5	12.6	7.1	2.4	(25)
(26)		MCWB	0.4	1.0	3.5	10.3	15.5	18.2	19.8	18.2	14.6	10.6	6.2	1.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.4	9.2	14.7	21.1	22.2	23.4	22.9	19.1	15.4	11.5	7.5	(27)
(28)		MADB	5.1	5.2	13.9	20.5	27.1	27.6	29.4	28.7	25.2	18.6	13.1	8.1	(28)	
(29)		2%	WB	2.6	2.8	6.7	13.1	18.8	20.7	22.0	21.3	17.4	13.8	9.3	5.3	(29)
(30)		MADB	3.1	3.4	10.2	18.2	24.5	26.2	27.5	27.3	22.9	17.0	10.1	5.9	(30)	
(31)		5%	WB	1.5	1.9	5.2	12.0	17.4	19.7	21.1	20.1	16.2	12.2	7.7	3.3	(31)
(32)		MADB	2.0	2.5	7.6	16.9	23.2	25.0	26.4	26.5	20.5	14.3	8.3	3.7	(32)	
(33)		10%	WB	0.5	1.1	3.7	10.9	16.2	18.7	20.3	19.0	15.1	10.9	6.2	1.9	(33)
(34)		MADB	0.9	1.6	5.7	15.4	21.6	23.4	25.0	24.2	18.6	12.4	7.0	2.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	4.9	6.0	7.1	9.5	10.7	10.0	10.1	11.1	9.5	7.2	4.4	4.3	(35)	
(36)		5% DB	MADB	3.5	4.9	10.2	13.2	13.6	12.7	13.2	14.8	13.7	10.7	5.8	4.3	(36)
(37)		MCWBR	3.4	4.3	6.1	6.7	6.1	5.9	5.7	6.3	6.9	7.0	4.7	4.0	(37)	
(38)		5% WB	MADB	3.4	4.5	9.3	11.1	12.4	11.3	11.4	12.8	11.9	9.8	5.2	4.2	(38)
(39)	MCWBR	3.3	4.3	6.1	6.2	6.2	5.8	5.9	6.2	6.6	6.9	4.5	4.0	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.332	0.359	0.428	0.398	0.385	0.420	0.425	0.389	0.357	0.336	0.304	(40)	
(41)		taud	2.138	2.069	2.138	2.129	2.303	2.361	2.279	2.248	2.313	2.367	2.259	2.178	(41)	
(42)		Ebn at Noon	661	746	815	801	853	868	827	797	783	731	619	582	(42)	
(43)		Edh at Noon	72	104	120	138	122	116	124	121	101	78	65	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.71	3.11	4.12	5.56	5.86	5.57	4.82	3.19	1.63	0.66	0.48	(44)	
(45)		RadStd	0.15	0.26	0.47	0.38	0.41	0.51	0.42	0.47	0.35	0.29	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.63	N/A	N/A	+0.94	N/A	N/A	-142	-235	+103	N/A
(47)	Regional (9 neighbors)	+0.68	N/A	N/A	+0.90	N/A	N/A	-140	-212	+105	+41

Nomenclature: See separate page